Enhancing Cardiovascular Clinical Care Through Bio-Physics & Data-Driven Modelling

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Leveraging existing clinical infrastructure for a comprehensive vascular health assessment

- Calibration-free estimation of arterial stiffness index β
- Calibration-free estimation of BP
- Pulse Wave Reflection Analysis
- Vessel Diameter & Blood Blow Velocity
- Wall Shear Stress
- Oscillatory Shear Index

Duplex Mode Ultrasonography

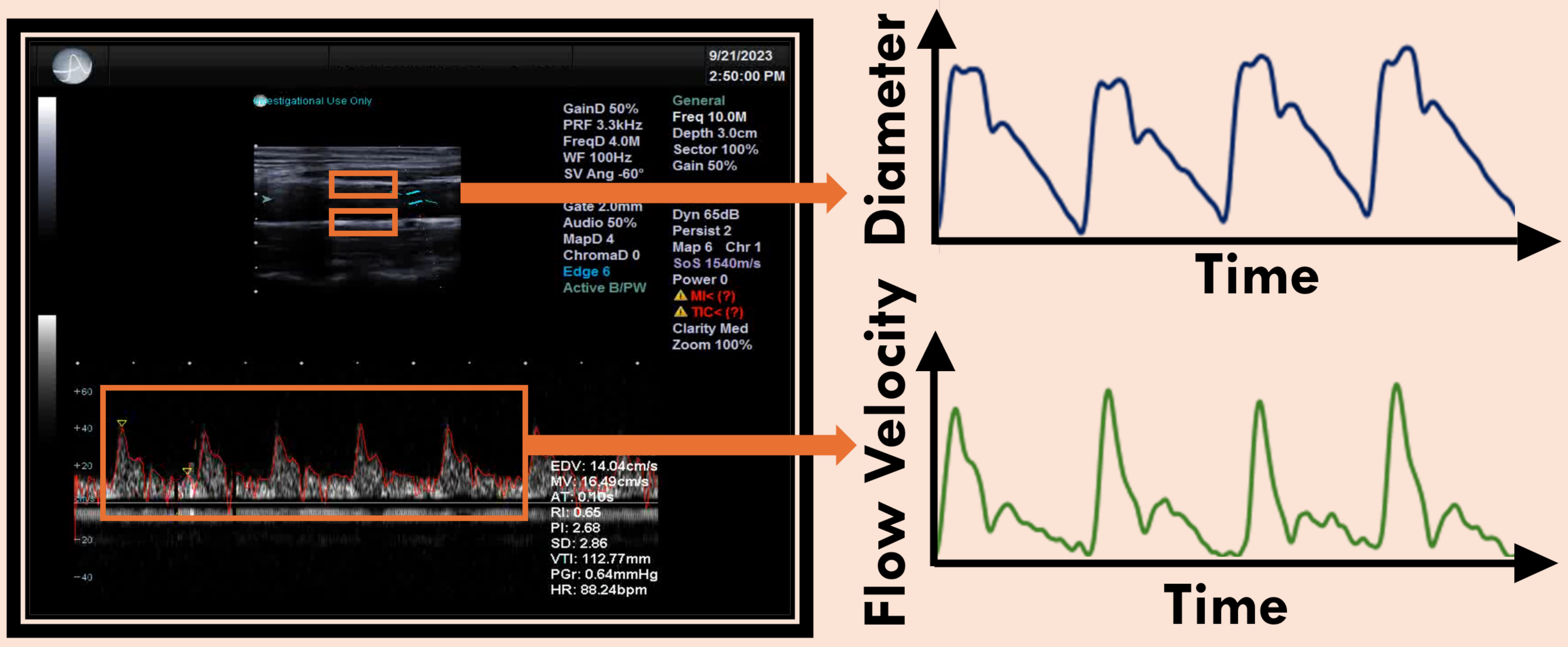
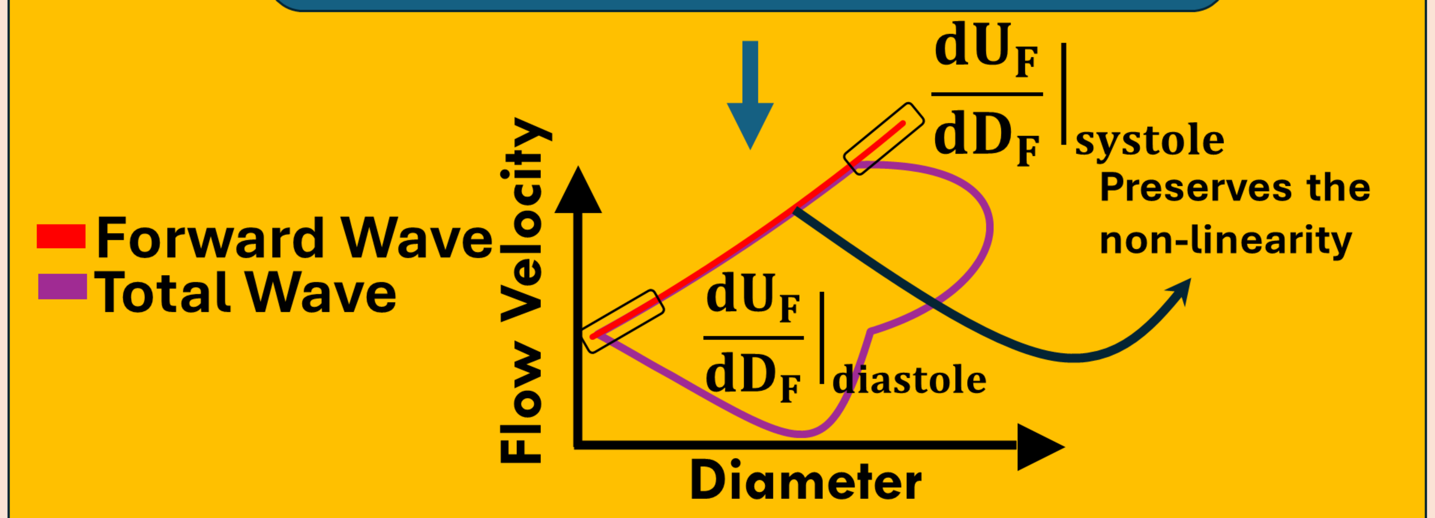


Image Processing to extract the pulse waveforms

Estimation of arterial stiffness index β

Diameter → Flow Velocity

Non-Linear Wave Separation (Cepstral Analysis)



$$\beta = f\left(\frac{dU_F}{dD_F}\Big|_{systole}, \frac{dU_F}{dD_F}\Big|_{diastole}, \text{Diameter}\right)$$

Digital Twin Model for Examining the Left Ventricular Dysfunction using Shear Wave Elastography

- Heart Failure ← Increased Left Ventricular (LV) Dysfunction ← Higher LV Filling Pressure
- Un-met clinical need to non-invasively estimate LV Filling Pressure
- Ultrasound Shear Wave Elastography (SWE) Velocities are associated with the Myocardial Stiffness
- How to estimate LV Filling Pressure from SWE Velocities, accounting for confounding factors as well ?

Bio-Physics Finite Element Model of Heart Wall (Septum)

1

- Wall Anisotropy | Thickness
- Curvature | Viscosity
- Loading Conditions: Pre-load & After Load
- Age-dependent stiffness variability
- Noisy Data

} Ground Truth Synthetic Data

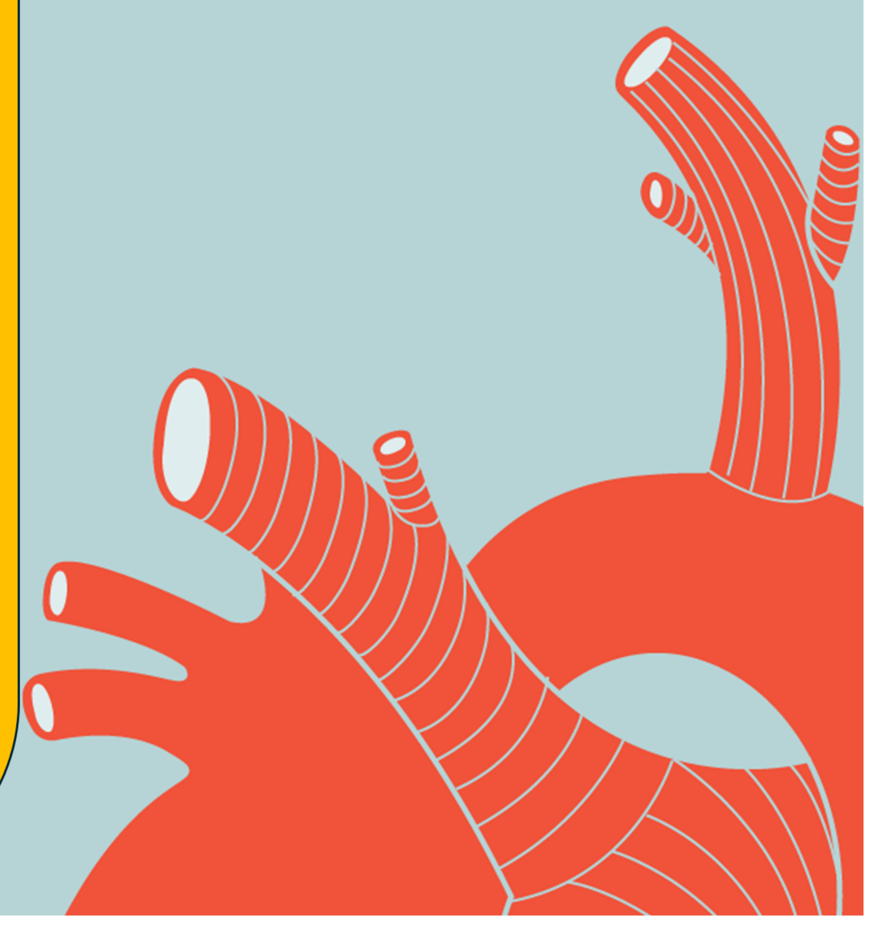
2

SWE Velocities + confounding factors

Digital Twin

LV Filling Pressure

✓ Validation & Testing with invasive data



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